

SOMALA AJAY

AI/ML Engineer | Gen AI Engineer | Python Developer

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PROFESSIONAL SUMMARY

AI/ML & Generative AI Engineer with 3+ years of experience developing production-grade LLM applications, RAG pipelines, and full-stack AI products. Strong foundation in data engineering, data validation workflows, and backend API integration. Proven track record of architecting scalable SaaS platforms, deploying vector search architectures, and delivering 20+ ML/DL models across NLP and Computer Vision. Proficient in Python, SQL, FastAPI, LangChain, FAISS, Docker, and Cloud (AWS/GCP/DigitalOcean).

WORK EXPERIENCE

TCS, Associate Software Engineer

12/2025 – 05/2026 | Hyderabad

- Designed and implemented ML pipelines for structured client datasets covering data ingestion, preprocessing, and model training.
- Optimized SQL queries and database schemas to handle large-scale data extraction, improving preprocessing data ingestion speeds by 20%.
- Performed EDA, data preprocessing, and feature engineering using Pandas and NumPy on datasets ranging from 10K to 50K records.
- Built classification models using Scikit-learn, achieving ~82–87% accuracy on validation datasets. Deployed ML model via Flask REST API for internal inference workflows.
- Automated data preprocessing pipeline reducing manual effort by ~25–30%. •Used SQL for data extraction and transformation to prepare model-ready datasets.
- Tracked and versioned ML experiments using Git, ensuring reproducibility across development iterations.

Tech Mahindra, Associate

09/2024 – 11/2025 | Hyderabad

- Worked on the client project using the Workflow Management System (WFMS) to handle and process structured data.
- Engineered high-volume unstructured and semi-structured text data pipelines within the Workflow Management System (WFMS) to prepare high-fidelity inputs for training downstream NLP and LLM architectures.
- Performed doc auditing and data correction to maintain high accuracy and data integrity.
- Supported data validation and quality checks, reducing discrepancies and improving workflow reliability.
- Designed and implemented robust data validation algorithms and quality verification workflows, eliminating label noise and reducing feature discrepancies by 20%.

BO IT Solutions Pvt Ltd, AI/ML Data Scientist

08/2022 – 06/2024 | Vijaywada

- Delivered 20+ end-to-end AI/ML projects across NLP, Computer Vision, and Deep Learning, covering the full pipeline from data collection to model deployment
- Performed EDA and feature engineering on datasets of 10K–100K+ records using Pandas, NumPy, and Matplotlib, reducing data preprocessing time by ~30%
- Built and evaluated classification and regression models using Scikit-learn and TensorFlow, consistently achieving 80–90% accuracy across multiple project deliveries
- Implemented CNN and LSTM-based deep learning models for image classification and sequence prediction tasks, improving baseline accuracy by ~15% over traditional ML approaches.

EDUCATION

B.Tech, Aditya University

2018 – 2022 | Surampalem, Andhra Pradesh

Electronics And Communication Engineering

TECHNICAL SKILLS

Generative AI & Agents — LLMs, RAG Pipelines, Semantic Search, Vector Databases, LangChain, LlamaIndex, Prompt Engineering, Agentic Workflows, ADK & MCP

Programming & Backend — Python, SQL, FastAPI, Flask, REST APIs

Machine Learning & DL — Scikit-learn, TensorFlow, Keras, CNN, LSTM, NLP, Feature Engineering, Model Evaluation

Data Engineering & DBs — Data Validation, Data Preprocessing, ETL Pipelines, SQL, Pandas, NumPy, MongoDB, MySQL,

Cloud & Developer Tools — Docker, Git, VS Code, GitHub Copilot, GCP, AWS, DigitalOcean

HACKATHON PARTICIPATION

National AI Hackathon - i2e Consulting, *Project: Complex Technical Manual QA System using RAG* Bangalore

- Built a production-grade RAG system ingesting and indexing 885 semantic vectors from the 270-page NASA Systems Engineering Handbook using FAISS and sentence-transformers (all-MiniLM-L6-v2)
- Integrated Groq API (LLaMA-3.3-70B) with confidence scoring (0–100%), multi-turn conversation memory, section-level citations, and NASA acronym expansion (20+ terms: TRL, PDR, KDP, SRR)

Stack: FastAPI | FAISS Groq API | LLAMA 3.3-70B | sentence-transformers | PyPDF | Vanilla JS | Python | PdfPlumber

KEY PROJECTS

PinGuru — Instagram DM Automation SaaS,

Python | FastAPI | MongoDB | Instagram Graph API | DigitalOcean | Stripe | GitHub Pages

- Architected and deployed **Pinguru**, a full-stack Instagram automation SaaS (pinguru.me) featuring real-time DM triggers and multi-tier Stripe billing. Built with a **FastAPI/MongoDB** backend and **Instagram Graph API**, the platform integrates robust security (JWT, OTP) and automated comment-to-DM workflows.

Complex Technical Manual QA System Using RAG, i2e-Consulting

https://github.com/Ajaysomala/i2e_Hireathon-Complex_Technical_Manual_QA_System 

- NASA Manual QA System | RAG Pipeline, Python | FastAPI | FAISS | Llama 3.3 | Groq | NLP. Designed a production-grade RAG pipeline using **FastAPI**, **FAISS**, and **Llama 3.3** to automate citation-backed QA for the 270-page NASA Systems Engineering Handbook.

Image Caption Generator, Python | CNN | LSTM | Deep Learning | NLP | FLICKR DataSet

- Built an end-to-end image captioning system using a CNN encoder for visual feature extraction and an LSTM decoder for sequential caption generation, trained on the Flickr8k dataset

Autonomous Multi-Agent Operations System,

Python | FastAPI | Groq API (Llama 3.3) | Multi-Agent Orchestration | Docker | SQLite | Pydantic

- Built a multi-layered AI agent system with increasing autonomy. A reflex layer handles known triggers with fixed rules. A single AI agent (Llama 3.3) interprets plain-language instructions and plans its own steps. For complex goals, a team of agents — Planner, Executor, and Reviewer — split the work, execute it, and self-correct through review. The whole system runs as a containerized FastAPI service with task memory.

CERTIFICATIONS

Generative AI Introduction

Microsoft Learning

Career Essentials in Generative AI by Microsoft and LinkedIn

Microsoft

Develop an AI app with the Azure AI Foundry SDK

Microsoft

Learning Microsoft 365 Copilot for Work

LinkedIn